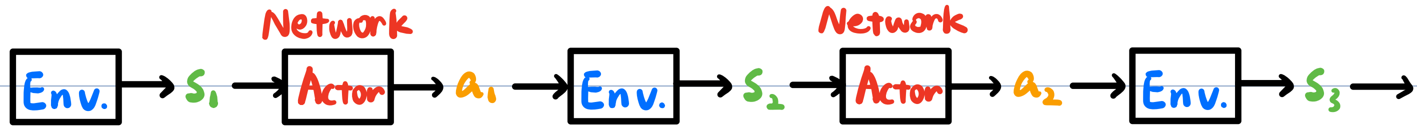
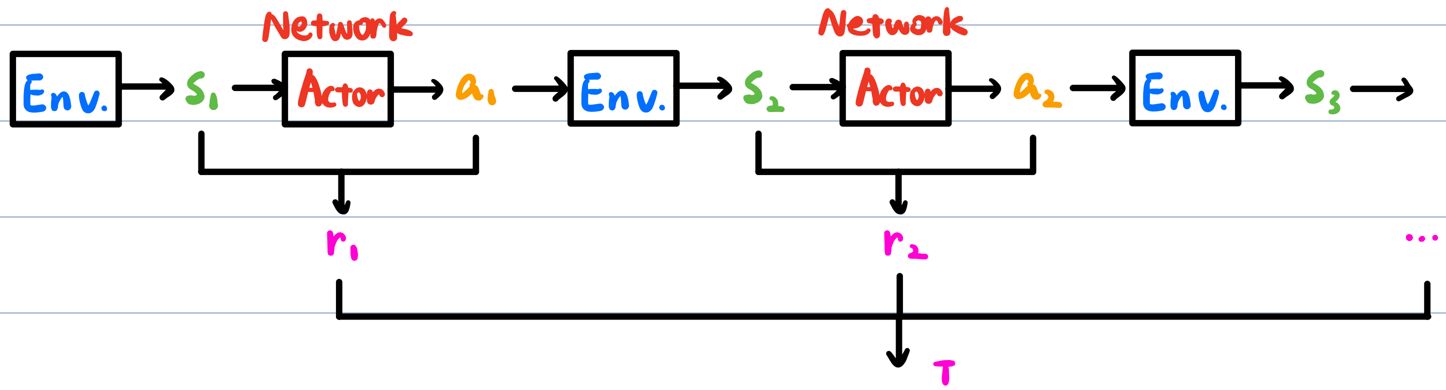
一、前置条件



轨迹 (Trajectory) $\tau = \{s_1, a_1, s_2, a_2, \cdots, s_T, a_T\}$

$$p_\theta(\tau) = p(s_1)\, p_\theta(a_1|s_1)\, p(s_2|s_1,a_1)\, p_\theta(a_2|s_2)\, p(s_3|s_2,a_2)\cdots$$
$$= p(s_1)\prod_{t=1}^{T} p_\theta(a_t|s_t)\, p(s_{t+1}|s_t,a_t)$$



$$R(\tau) = \sum_{t=1}^{T} r_t$$

Expected Reward : $\bar{R}_\theta = \sum_\tau R(\tau)\, p_\theta(\tau) = E_{\tau\sim p_\theta(\tau)}[R(\tau)]$

$\nabla f(x) = f(x)\nabla \log f(x)$

计算梯度

Policy Gradient : ① $\nabla \bar{R}_\theta = \sum_\tau R(\tau)\nabla p_\theta(\tau) = \sum_\tau R(\tau)\, p_\theta(\tau)\frac{\nabla p_\theta(\tau)}{p_\theta(\tau)}$

$$= \sum_\tau R(\tau)\, p_\theta(\tau)\, \nabla \log p_\theta(\tau)$$

$$= E_{\tau\sim p_\theta(\tau)}[R(\tau)\nabla \log p_\theta(\tau)]$$

$$\approx \frac{1}{N}\sum_{n=1}^{N} R(\tau^n)\nabla \log p_\theta(\tau^n)$$

$$= \frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T_n} R(\tau^n)\nabla \log p_\theta(a_t^n|s_t^n)$$

② $\theta \leftarrow \theta + \eta \nabla \bar{R}_\theta$

训练资料： 给定 Policy π_θ

τ^1: (s_1^1, a_1^1) $R(\tau^1)$

(s_2^1, a_2^1) $R(\tau^1)$

… …

τ^2: (s_1^2, a_1^2) $R(\tau^2)$

(s_2^2, a_2^2) $R(\tau^2)$

… …

二、On-policy v.s. Off-policy

On-policy：用于训练的 Actor 和用于交互的 Actor是同一个

Off-policy：用于训练的 Actor 和用于交互的 Actor是不一样的

1、On-policy → Off-policy

$$\nabla \bar{R}_\theta = E_{\tau \sim p_\theta(\tau)}[R(\tau) \nabla \log p_\theta(\tau)]$$

①利用策略 π_θ 收集数据，当 θ 更新时，需重新采集数据

②新的目标：从 $\pi_{\theta'}$ 中采集数据训练 θ。θ' 是固定的，所以重复使用采样数据

重要性采样： $E_{x\sim p}[f(x)] \approx \cancel{\frac{1}{N}\sum_{i=1}^{N} f(x^i)}$ x^i 是从 $p(x)$ 中采样得到

$$= \int f(x)\, p(x)\, dx$$

$$= \int f(x) \frac{p(x)}{q(x)} q(x)\, dx$$

x^i 只能从 $q(x)$ 中采样得到.

$$= E_{x\sim q}[f(x)\frac{p(x)}{q(x)}]$$

$$Var_{x\sim p}[f(x)] = E_{x\sim p}[f(x)^2] - (E_{x\sim p}[f(x)])^2$$

$$Var_{x\sim q}[f(x)\frac{p(x)}{q(x)}]$$

$$= E_{x\sim q}[(f(x)\frac{p(x)}{q(x)})^2] - (E_{x\sim q}[f(x)\frac{p(x)}{q(x)}])^2$$

$$= E_{x\sim p}[f(x)^2\frac{p(x)}{q(x)}] - (E_{x\sim p}[f(x)])^2$$

$$\nabla \bar{R}_\theta = E_{\tau\sim p_{\theta'}(\tau)}\left[\frac{p_\theta(\tau)}{p_{\theta'}(\tau)} R(\tau)\nabla \log p_\theta(\tau)\right]$$

① 从 θ' 中采样数据

② 利用采集到的数据多次训练 θ

要更新的梯度 $\nabla J^{\theta'}(\theta)$ 对 θ 的梯度

$$= E_{(s_t,a_t)\sim\pi_\theta}[A^\theta(s_t,a_t)\nabla \log p_\theta(a_t^n|s_t^n)]$$

$$= E_{(s_t,a_t)\sim\pi_{\theta'}}\left[\frac{p_\theta(s_t,a_t)}{p_{\theta'}(s_t,a_t)} \overset{A^{\theta'}(s_t,a_t)}{\cancel{A^{\boxed{\theta}}(s_t,a_t)}}\nabla \log p_\theta(a_t^n|s_t^n)\right]$$

设 $p_\theta(s_t) \approx p_{\theta'}(s_t)$

$$= E_{(s_t,a_t)\sim\pi_{\theta'}}\left[\frac{p_\theta(a_t|s_t)}{p_{\theta'}(a_t|s_t)} \cancel{\frac{p_\theta(s_t)}{p_{\theta'}(s_t)}} A^{\theta'}(s_t,a_t)\nabla \log p_\theta(a_t^n|s_t^n)\right]$$

$$p_\theta(a_t|s_t)\nabla \log p_\theta(a_t|s_t) = \nabla p_\theta(a_t|s_t)$$

$$J^{\theta'}(\theta) = E_{(s_t,a_t)\sim\pi_{\theta'}}\left[\frac{p_\theta(s_t|a_t)}{p_{\theta'}(s_t|a_t)} A^{\theta'}(s_t,a_t)\right]$$

$\theta \longleftarrow \theta + \alpha \nabla_\theta J(\theta)$

三、PPO (Proximal Policy Optimization)

$$J^{\theta'}_{PPO}(\theta) = J^{\theta'}(\theta) - \underline{\beta KL(\theta, \theta')}$$ ⟵ 对θ和θ'输出的Action做KL

期望θ与θ'越像越好

(1) PPO 算法

① 初始化策略参数 θ^0

② 对于每次迭代

1. 使用 θ^k 与环境交互得到 $\{s_t, a_t\}$ 并计算 advantage $A^{\theta^k}(s_t, a_t)$

2. 找到 θ 以优化 $J_{PPO}(\theta)$

$$J^{\theta^k}_{PPO}(\theta) = J^{\theta^k}(\theta) - \beta KL(\theta, \theta^k)$$ 多次更新参数

其中 $J^{\theta^k}(\theta) \approx \sum_{(s_t, a_t)} \frac{P_\theta(a_t | s_t)}{P_{\theta^k}(a_t | s_t)} A^{\theta^k}(s_t, a_t)$

如果 $KL(\theta, \theta^k) > KL_{max}$，增大 β

如果 $KL(\theta, \theta^k) < KL_{min}$，减小 β

自适应KL惩罚

(2) PPO 2 算法

$$J^{\theta^k}_{PPO2}(\theta) = \sum_{(s_t, a_t)} \min\left(\frac{P_\theta(a_t | s_t)}{P_{\theta^k}(a_t | s_t)} A^{\theta^k}(s_t, a_t),\right.$$

$$\text{clip}\left(\frac{p_\theta(a_t|s_t)}{p_{\theta^k}(a_t|s_t)}, 1-\varepsilon, 1+\varepsilon\right)A^{\theta^k}(s_t, a_t))$$